

ELIoT-RTU

User Manual

Emmanuel's Lanka (Pvt) Ltd.

Warnings

This device should be installed by authorized personal only



Avoid handling this product while it is powered.

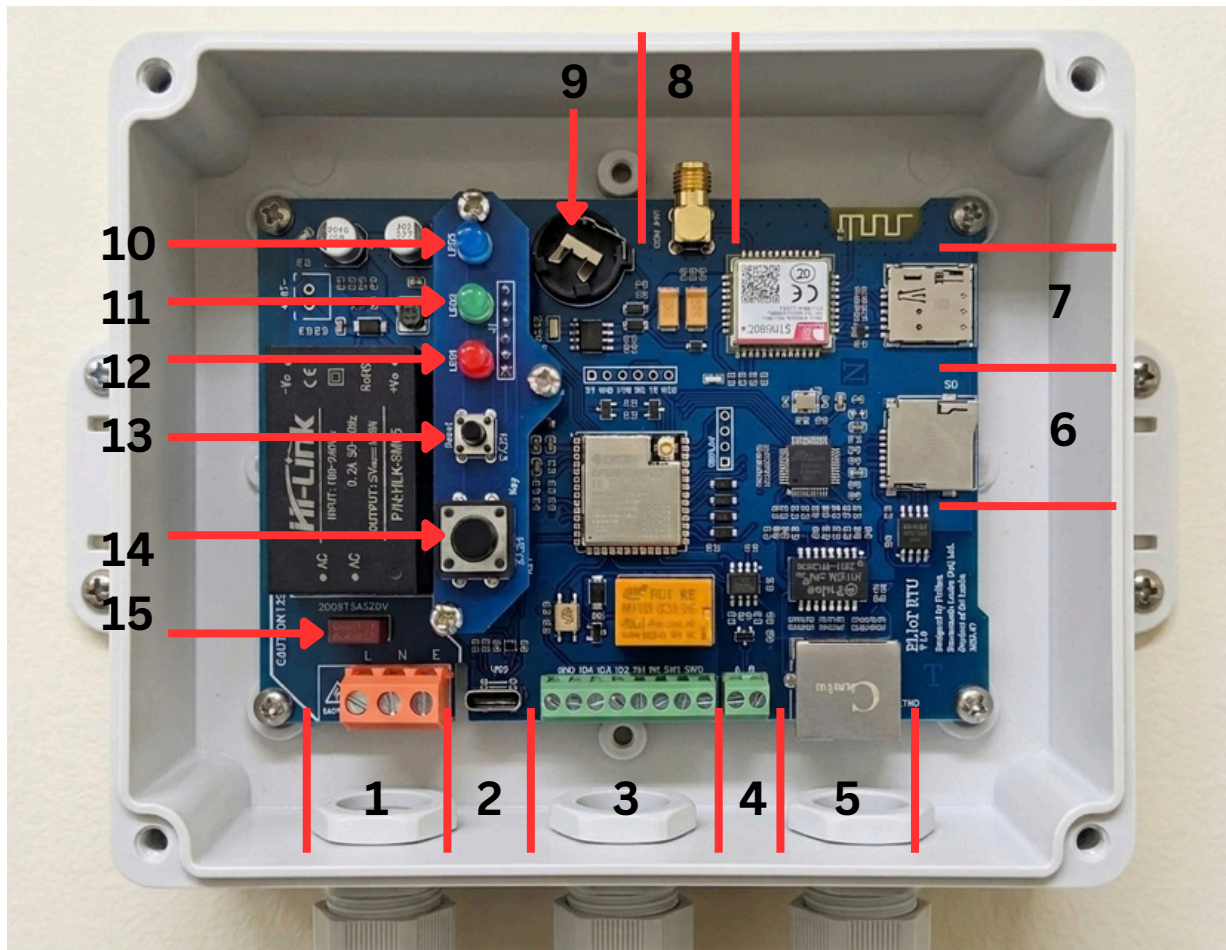
Do not expose this product to heat from any source.

Take care while handling this product to avoid mechanical or electrical damage to the printed circuit board and connectors.

Any peripheral or equipment used with the RTU-device should comply with relevant standards for the country of use and be marked accordingly to ensure that safety and performance requirements are met.

Do not leave the power supply connected during lightning activity.

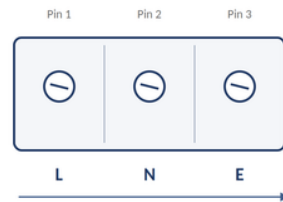
Components diagram



1. Mains Power Connector
2. USB port
3. GPIO port
4. RS485 port
5. Ethernet port
6. SD card slot
7. SIM slot
8. GSM antenna connector
9. RTC Battery holder
10. System status LED
11. Network status LED
12. Fault status LED
13. Reset switch
14. General purpose button
15. Fuse

1. Mains Power Connector

3-pin terminal block (Live, Neutral, Earth) for connecting mains power to the unit.



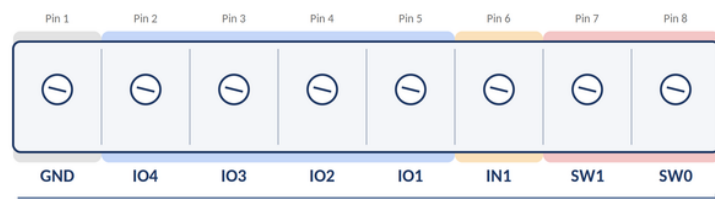
2. USB Port

USB OTG; used for local connection to a PC for diagnostics and firmware access.

3. GPIO Port

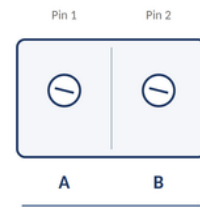
8-pin terminal block (GND, IO4–IO1, IN1, SW1, SW0):

- GND – Ground reference
- IO4–IO1 – Four general-purpose I/O pins, 3.3 V logic level
- IN1 – Opto-isolated input, 5 V signal
- SW1, SW0 – Relay output contacts (normally-open), rated 5 A @ 30 V DC / 3 A @ 250 V AC



4. RS-485 Port

2-pin terminal block (A, B) for connecting the device to an inverter, meter or sensor.



5. Ethernet Port

RJ45 connector providing a 10/100 Mbps wired network connection.

6. SD Card Slot

Support microSD card up to 32 GB (not included).

7. SIM Slot

Accepts a Nano SIM.

8. GSM Antenna Connector

SMA connector for external 4G antenna.

9. RTC Battery Holder

Use CR1220, 3 V lithium cell.

10. System Status LED

Indicates the system is active; shows a breathing pattern and blinks at 1 Hz during active operation.

11. Network Status LED

Indicates cellular network status – static: network OK, blinking: no SIM inserted, off: no network connection.

12. Fault Status LED

Indicates a fault condition, such as a communication error, abnormal communication, or an unknown device state.

13. Reset Switch

Restarts the device.

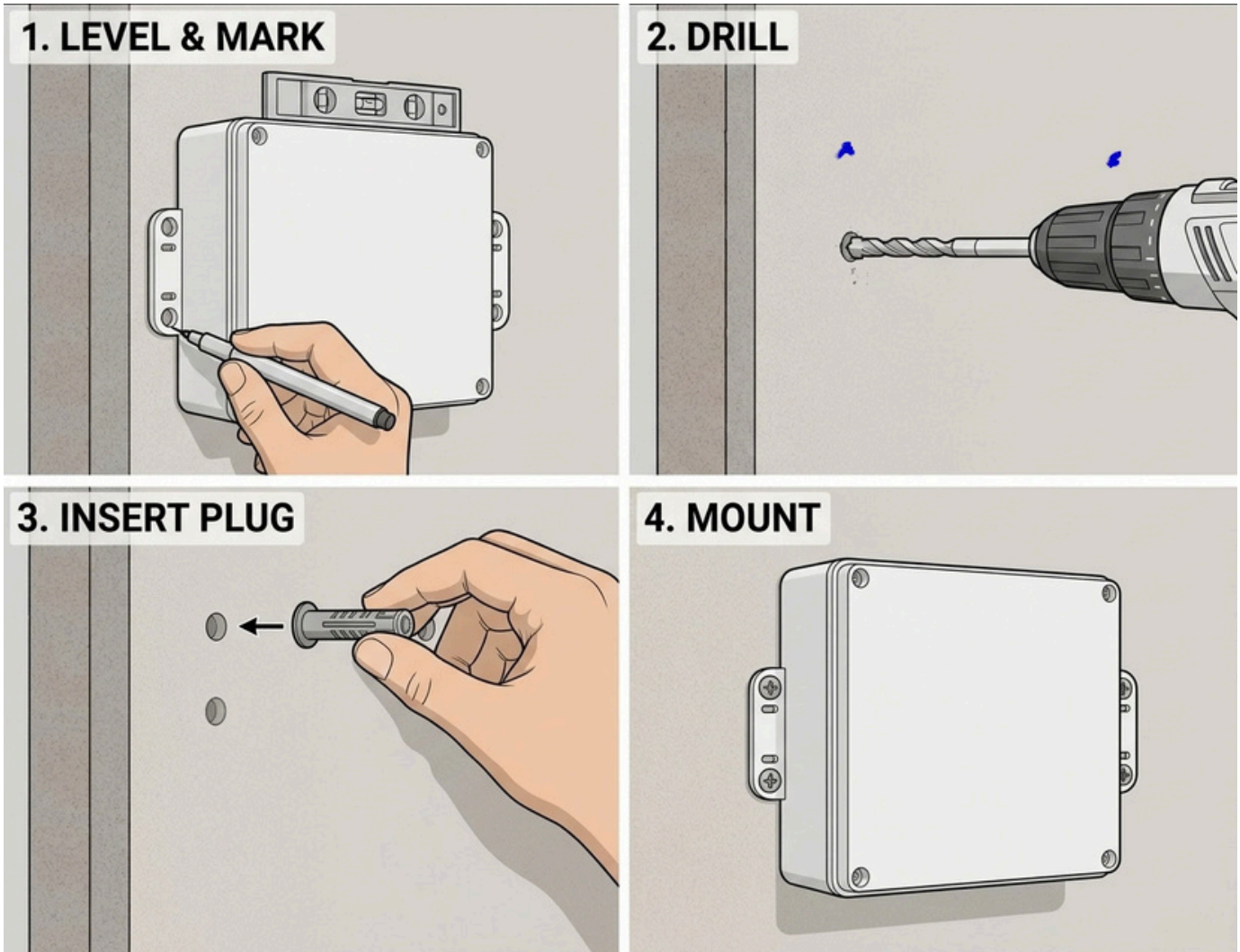
14. General Purpose Button

User-assignable button for mode selection or triggering user-defined actions.

15. Fuse

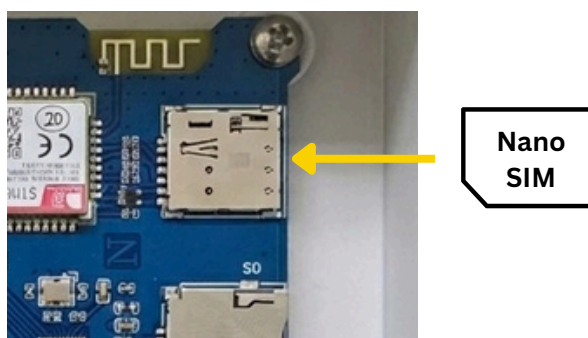
250 V, 5 A fuse protection, Replace only with part 2009T5A250V.

Installation Procedure



Screw : 5 × 70 mm countersunk
Head : Countersunk (CSK)
Wall plug : 8 × 60 mm
Drill bit. : 8 mm masonry bit

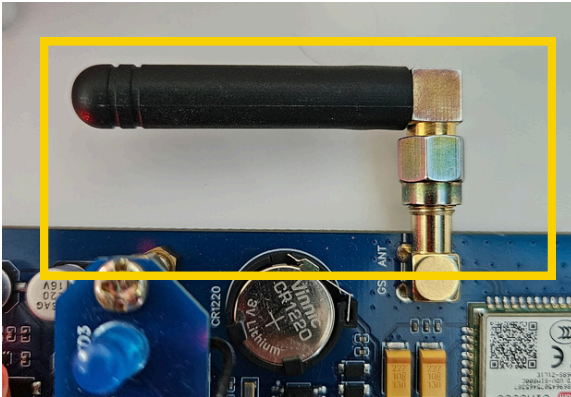
5. Insert SIM card



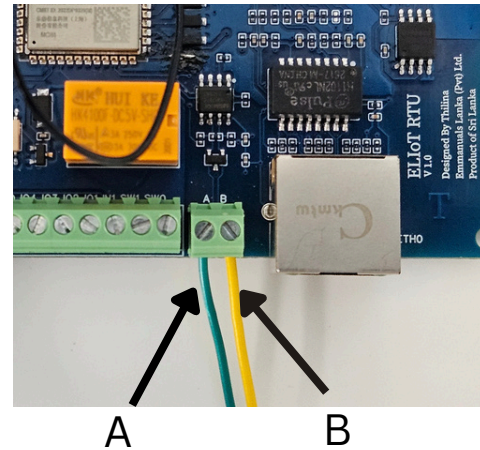
6. Insert Battery



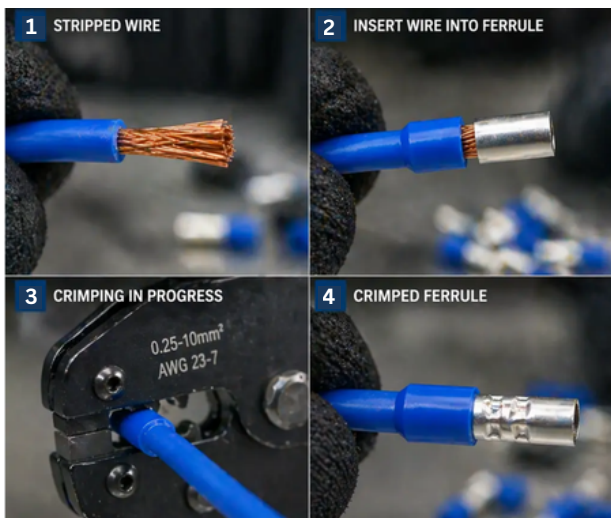
7. Connect 4G antenna



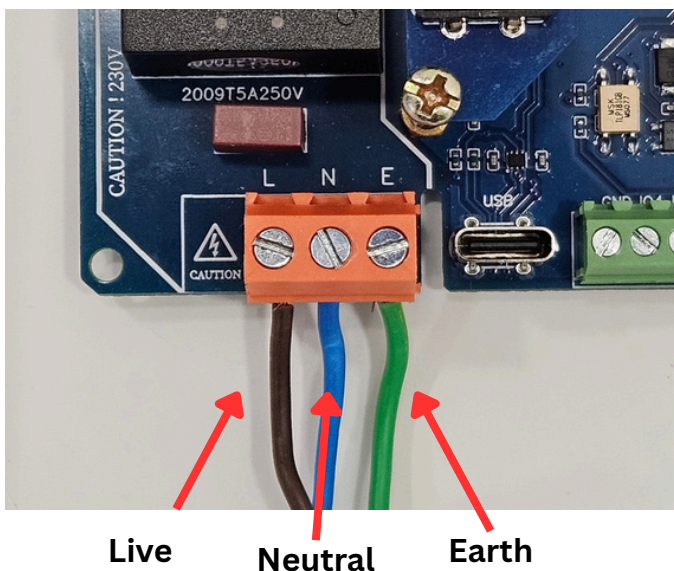
8. Connect RS485 cable



9. Prepare Power Cable



10. Insert the Power Cable



Device Setup

1. Switch On the Device
2. Wait about 2 mins to connect to the network
3. Open Wi-Fi settings and connect to ELIoT-RTU.
4. Enter the provided password.
5. Your device may say No Internet. This is normal. Stay connected.
6. Open a web browser and go to <http://192.168.4.1>, or scan the QR code below after connecting.



7. Follow the on screen instructions for further configurations.